Assignment 6.4

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Task 1

```
# Student Performance Evaluation System

class Student:
    def __init__(self, name, roll_number, marks):
        self.name = name
        self.roll_number = roll_number
        self.marks = marks

    def display_details(self):
        print("Student Name:", self.name)
        print("Roll Number:", self.roll_number)
        print("Marks:", self.marks)

    def check_performance(self, class_average):
        if self.marks > class_average:
            return "Above class average"
        else:
            return "Below class average"
```

```python
# Example usage
student1 = Student("Kiran", "101", 82)

student1.display_details()

avg = 75
print("Performance:", student1.check_performance(avg))
```

Output

```
Student Name: Kiran
Roll Number: 101
Marks: 82
Performance: Above class average
```

Task 2

```python
# Sensor readings list
readings = [12, 7, 18, 5, 20, 9]

for value in readings:
    if value % 2 == 0:
        square = value ** 2
        print(f"Even reading: {value}, Square: {square}")
```

Output

```
Even reading: 12, Square: 144
Even reading: 18, Square: 324
Even reading: 20, Square: 400
```

Task 3

```
# Banking Transaction Simulation

class BankAccount:
    def __init__(self, account_holder, balance):
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print(f"Deposited: {amount}")
        print(f"New Balance: {self.balance}")

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print(f"Withdrawn: {amount}")
```

```
            print(f"Remaining Balance: {self.balance}")
        else:
            print("Insufficient balance. Withdrawal denied.")


# Example usage
account = BankAccount("Ravi", 5000)

account.deposit(1000)
account.withdraw(2000)
account.withdraw(6000)
```

Output

```
Deposited: 1000
New Balance: 6000
Withdrawn: 2000
Remaining Balance: 4000
Insufficient balance. Withdrawal denied.
```

Task 4

```
# Copilot: write a while loop that iterates through the students list
```

and prints names of students whose score is greater than
75

index = 0

while index < len(students):
    if students[index]["score"] > 75:
        print("Eligible:", students[index]["name"])
    index += 1

Output

Eligible: Ravi
Eligible: Kiran
Eligible: Arjun

Task 5

Online Shopping Cart Module

class ShoppingCart:

```
    def __init__(self):
        self.items = []

    def add_item(self, name, price, quantity):
        item = {"name": name, "price": price,
"quantity": quantity}
        self.items.append(item)
        print(f"Added {name} to cart.")

    def remove_item(self, name):
        for item in self.items:
            if item["name"] == name:
                self.items.remove(item)
                print(f"Removed {name} from cart.")
                return
        print("Item not found.")

    def calculate_total(self):
        total = 0

        for item in self.items:
            total += item["price"] * item["quantity"]
```

```python
        # Apply discount
        if total > 1000:
            discount = total * 0.1
            total -= discount
            print(f"Discount applied: {discount}")

        return total


# Example usage
cart = ShoppingCart()

cart.add_item("Shoes", 800, 1)
cart.add_item("T-shirt", 300, 2)

cart.remove_item("Shoes")

print("Total Bill:", cart.calculate_total())
```

Output

```
Added Shoes to cart.
Added T-shirt to cart.
Removed Shoes from cart.
Total Bill: 600
```